Derivation of the Normal Equation

Linear Regression – Least Squares (Closed Form)

Linear Regression Model

- We assume a linear model: $\hat{y} = X\theta$
  - $X \in \mathbb{R}$^{m×n}: design matrix (features + bias column)
  - $\theta \in \mathbb{R}$^{n×1}: parameter (weight) vector
  - $y \in \mathbb{R}$^{m×1}: target values

Loss Function (Least Squares)

- Goal: minimize prediction error
  - $J(\theta) = (1/2)\ ||X\theta - y||^2$
  - Squared Euclidean norm of residuals
  - Factor 1/2 simplifies differentiation

Matrix Form of the Loss

- $||v||^2 = v^T v$ for any vector v
  - $J(\theta) = (1/2)(X\theta - y)^T(X\theta - y)$
  - Ensures the loss is a scalar

Expand the Quadratic Form

- $J(\theta) = (1/2)[\ \theta^T X^T X\theta - 2y^T X\theta + y^T y\ ]$
  - Uses transpose and distributive properties
  - $X^T X$ is symmetric

Gradient with Respect to θ

- Take derivative of $J(\theta)$ w.r.t. θ
  - $\nabla_\theta J(\theta) = X^T X\theta - X^T y$
  - Gradient has same shape as θ

Set Gradient to Zero

- At minimum: $\nabla \theta \, J(\theta) = 0$
  - $X^T X\theta - X^T y = 0$
  - Rearranged: $X^T X\theta = X^T y$

Normal Equation (Closed Form Solution)

- If $X^T X$ is invertible:
  - $\theta = (X^T X)^{-1} X^T y$
  - Gives exact least-squares solution

Geometric Interpretation

- Solution projects y onto column space of X
  - Residual vector is orthogonal to X
  - Equivalent to minimizing squared distances

Practical Considerations

- Requires $X^T X$ to be invertible (full rank)
  - Numerical instability for large or ill-conditioned X
  - Gradient descent often preferred in practice